



معلومات جينية - علم الوراثة الجزيئية

MOLECULAR GENETICS

**THE BOOK:
FROM GENE TO
GENOME**

**CHAPTER 1
GENETICS : THE STUDY OF
BIOLOGICAL INFORMATION**

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Chapter 1: The Study of Biological Information.

Concept 1.1 – DNA: Life's Fundamental Information Molecule.

As we know, the DNA stores the biological information in what's called nucleotides (A – T – C – G) also we ALL know that the DNA has the double stranded helix shape.

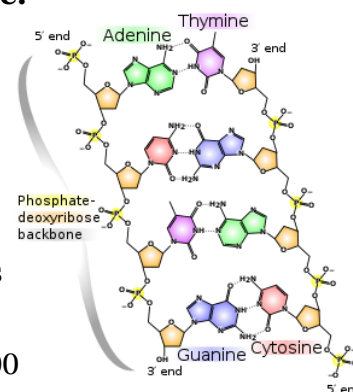
The bond between the complementary base pairs is a hydrogen bond, while between each base is a phosphodiester bond.

The DNA is in a 3D shape, but its info is 1D – it's encoded as sequences –, and digital – stored in computers –.

Genes is the DNA regions that encode proteins. Human cells have 27000 genes.

Genome is the entire collection of chromosomes in each cell.

Chromosome is an organized structure containing DNA and proteins that package and manage the storage, duplication, expression and evolution of DNA.



Concept 1.2 – Proteins: The Functional Molecules of Life Processes.

Most characteristics that a living organism has – metamorphism, moving, metabolism – are arose from proteins, which are large polymers of thousands of amino acids.

there are 20 amino acids that are translated from the m-RNA and make the protein. The differences in their orientation gives them the ability to make thousands of proteins.

the protein diversity of its 3D shape gives diversity in its function, as the shape of Haemoglobin and Lactate dehydrogenase differ, their function (oxygen transport & digest lactate) also differ.

Concept 1.3 – Molecular Similarities of All Life-Forms.

Many Biologist think that the RNA was the first information processing molecule to appear. RNA has 4 subunits (A- U – C – G), and it has the capacity to store, replicate and express information. And it can fold in a 3D shape and function like proteins.

but still, DNA is more stable than RNA, and that's why DNA stores information and replicates. And proteins have a greater diversity and contribute in a large part of functions more than the 3D RNA.

which makes the RNA in the end, the intermediate in between the DNA and the proteins.

RNA functions as an intermediate by the genetic code, in which each codon – 3 adjacent nucleotides – codes for an amino acid. This method is a huge evidence of a common ancestor for all living forms, since it's common for all.

Examples:

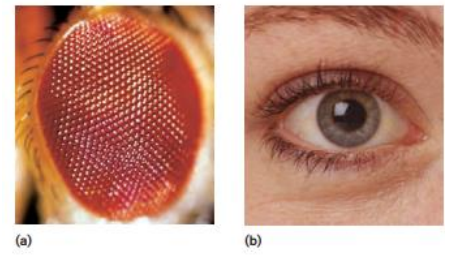
- The Cytochrome C protein is mostly identical in the diverse species, and the tiny differences of its amino acid is from the accumulative mutations that happened.
> don't forget that these mutations are a main force of evolution.

- The gene pax6 is important in eye development for all organisms.

A study was made by taking the human pax6 gene into a fruit fly genome, and the results shown a normally functioning fruit fly eye.

Which indicates that the gene functions the same, but its interactions with the genome it's in is what made a different eye between humans and fruit flies

Figure 1.9 The eyes of insects and humans have a common ancestor. (a) A fly eye and (b) human eye.



بالعربي: في جين مسؤول عن تكوين العين اسمه pax6 وهو موجود بكل الكائنات الحية تقريبا، صارت تجربة تم نقل فيها النسخة البشرية من الجين هاد لجينوم ذبابة فأكهة، وكانت النتيجة انه طلعت عين طبيعية لذبابة الفاكهة وما تأثرت من التغيير الي صار فيها. وهاد دليل انه كان في جين واحد مشترك تضاعف وصار فيه شوية طفرات أدت للاختلاف بين العيون.

Concept 1.4 – The Modular Construction of Genomes.

Gene families are made when a primordial gene duplicates then each copy of it has different mutations which gives them different functions. a common example is the rhodopsin gene in humans and chimpanzees.

The principle that underlies the evolution process is called Duplication followed by Divergence.

The protein coding gene has 10 or more exons – protein coding region in the gene – separated by introns – noncoding regions in the gene -.

About 3.7 billion years ago, prokaryotes like bacteria evolved. And about 2 billion years ago eukaryotes like algae evolved. Yet only 700-600 million years ago the multicellular eukaryotes appeared. Then 570 the Cambrian explosion happened, and the multicellular life-forms diverged.

Cambrian explosion: which is an event that took 20-50 million years that had the all major animal phyla started appearing.

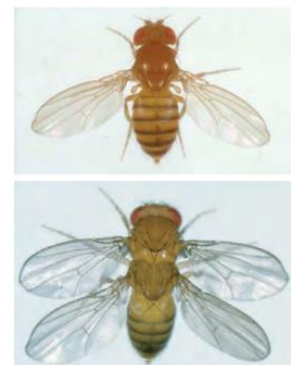
Q – how could the multicellular forms diverse that fast?

A- It's because the gene families sometimes rapidly expand to gene superfamilies containing hundreds of genes that have the same but slightly different function. example: the immune system is encoded by gene superfamilies in human adults and mice.

B- the rapid change of regulatory networks that specify how gene behave during development.

example: the two- winged fly evolved from a four- winged fly because of some changes in the regulatory networks. The 2 wings which were gone were replaced by balancing organs known as halteres.

Regulatory network: A gene regulatory network is a collection of molecular regulators that interact with each other and with other substances in the cell to control the gene expression levels of mRNA and proteins.



Concept 1.5 – Modern Genetic Techniques

Genetic Dissection of Model Organisms Reveals the Working of Biological Processes

The genetic dissection logic is about inactivating a gene in a model organism (Bacteria, yeast, nematodes, fruit flies, mice, zebra fish, Arabidopsis) and observe the consequences. like the loss of a gene for visual pigment in fruit fly produces a white eyed fly. This means that the lost gene had a role in eye pigmentation.

Whole-Genome Sequencing Can Identify Mutant Genes that Cause Disease.

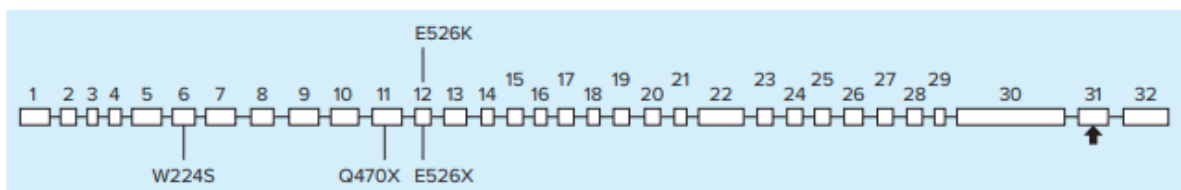
Genome sequencing technology is now used to identify the gene mutations that cause human genetics diseases such as microcephaly, a rare brain malformation.

This disease is known to happen when a mutation happens in a gene called WDR62, scientist had to sequence two kids with the same disease to indicate the mutation, afterwards several new mutations have been discovered that causes different forms of the disease. which concluded that the gene WDR62 is a causative gene for microcephaly.

بالعربي : في طفرة بالجين WDR62 يتسبب مشكلة بنمو الدماغ فبالنتالي ييجي حجمه غير طبيعي، العلماء اكتشفوا هاد المرض بعد ما شافوا جينوم حالتين من هاد المرض ولقوا عندهم نفس الطفرة في نفس الجين. وبعد فترة تبينت حالات ثانية لنفس المرض ولكن عندهم طفرة غير. بالتالي استنتجنا انه هاد الجين ممكن يسبب هاد المرض في حال تعرض لطفرة معينة.

In the figure below you can see letters and numbers like W224s, it means that the 224th amino acid is normally w (tryptophan) but it's changed to s (serine). the figure shows five different mutations that can cause the microcephaly disease.

(c) WDR62 gene mutations in different families with microcephalic children



Gene Therapy May Help Cure Genetic Diseases.

Since the sequencing of the whole genome is now known, it's possible to inform parents whether their children might suffer from a certain disease or not. this will help then consider ways to avoid/prepare for the outcome.

Also, identification of a disease gene gives information about the encoded protein that can guide us to design effective therapeutics as a treatment. This strategy worked in developing drugs that compact genetic diseases including certain kinds of cancer.

Gene therapy is one of the methods that are currently under development, where scientists introduce normal copies of genes into human cells. Genome editing is an alternative approach where researchers change the base-pair sequence of a mutant gene to its normal counterpart. Both methods have been used in mice to reverse a disease process or restore a gene function, but it's still too early to apply them on humans.

Concept 1.6 – Human Genetics and Society

It's good to know that some mutations directly cause the disease, such as the sickle cell disease which happens with a single change in a specific single DNA base pair in the B-globin gene.

but other mutations predispose the disease, such as breast cancer, which a mutation in BRCA1 gene increases the risk of breast cancer between 40%-80% in a woman carrying one copy of the mutation.

Many Social Issues Need to Be Addressed

The differences in each individual's genome can provide us information about susceptibility or resistance to many diseases as well as the potential for the expression of many physiological and neurological attributes. These differences may be known in less than a decade, and more issues will be solved about the aspects of a person's genetic make-up.

Issues :

- Society can restrict some lives because of its genome, so each individual genome should remain a protected privacy
- Detailed genetic profiles may give misinterpretation of some certain kind of information, which will be useless at best, and harmful at worst.
- Gene therapy technology may alter/add to the genes in the germ line cells of human embryos. The genetic engineering methods that's been mentioned earlier are used in non-inheritable somatic cells.

To prevent such issues just get educated =)